

BALASUBRAMANI P

AI & Machine Learning Engineer

Disease Prediction · NLP & Speech AI · Full-Stack ML Deployment



CONTACT

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Location

Coimbatore, Tamil Nadu

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AI / ML SKILLS

- Machine Learning (scikit-learn)
- Naive Bayes Classifier
- TF-IDF Vectorization
- NLP & Text Processing
- OpenAI Whisper (ASR)
- SpeechRecognition
- Python (NumPy, Pandas)

DEVELOPMENT

- FastAPI / RESTful APIs
- Streamlit
- PostgreSQL / SQLite
- SQLAlchemy ORM
- JWT Authentication
- Java
- HTML / CSS

TOOLS & IDES

- VS Code / Eclipse
- Git / GitHub
- Postman (API Testing)

EDUCATION

B.E. – ECE (2026)

V.S.B. College of Engineering CGPA: 7.8 | Minor: CMOS Design

HSC (2022)

HNUC Higher Secondary | 72.67%

SSLC (2020)

HNUC Higher Secondary | 78.8%

/ AI / ML OBJECTIVE

Final-year ECE engineer with hands-on AI/ML project experience, seeking roles in Machine Learning, AI Development, or Data Science. Proven ability to build end-to-end intelligent systems — from model training to full-stack deployment — with a strong focus on real-world healthcare and NLP applications.

/ PROJECTS

AI Healthcare System for Appointment Booking 2024 – 2025

Stack: FastAPI · Streamlit · PostgreSQL · scikit-learn · Naive Bayes · TF-IDF · OpenAI Whisper · JWT

- Engineered an AI-powered disease prediction engine using Multinomial Naive Bayes with TF-IDF vectorization, delivering top-3 diagnoses with confidence scores in real time.
- Built multi-modal symptom input — text, voice recording, and audio upload — powered by OpenAI Whisper and SpeechRecognition for seamless speech-to-text.
- Designed a full multi-role platform (Patient · Doctor · Admin) with JWT-based auth, dynamic slot availability, and doctor recommendation engine (location, specialization, rating).
- Architected RESTful backend with FastAPI and SQLAlchemy ORM on PostgreSQL; clean Streamlit frontend with role-specific dashboards, medical history, and prescription management.

AI Voice Assistant – Personal Productivity Tool Jun 2023 – Jul 2025

Stack: Python · SpeechRecognition · NLP · REST APIs

- Developed an offline-capable voice assistant in Python for task execution, reminders, and API-based web search.
- Integrated NLP pipelines for intent recognition, enabling natural language command execution and real-time information retrieval.

Medical & Portfolio Websites – Frontend Jun 2023 – Jul 2023

Stack: HTML · CSS

- Designed and deployed a responsive medical services website and a personal portfolio, handling full UI/UX implementation.

/ INTERNSHIP

Software Engineering Intern Mphasis Limited, Bengaluru | 2024

- Worked within an enterprise engineering team at Mphasis, gaining exposure to production-grade software development, agile workflows, and cross-functional collaboration.
- Applied problem-solving and code quality practices in a professional setting, reinforcing software engineering best practices.

/ KEY STRENGTHS FOR AI / ML ROLES

End-to-End ML Systems Building from data preprocessing to model training, API deployment, and UI integration.

CERTIFICATIONS

Generative AI for Educators Google · 2025
STEM-25 International Conference Jai
Shriram Engg. College · 2025
Electronic Systems Design (PCB)
NPTEL · 2024
Full Stack Web Development STARK
Institutions · 2024

NLP & Speech AI Hands-on experience with TF-IDF, Whisper ASR, and SpeechRecognition for production use.

Full-Stack Deployment FastAPI + Streamlit + PostgreSQL stack; JWT auth; RESTful API design.

Multi-Role System Design Architected complex multi-actor platforms with business logic, data models, and access control.

I hereby declare that all information provided is accurate to the best of my knowledge.

Place: Coimbatore | Date: 15-11-2025 | **Balasubramani P**